

MPV MEDIA PLAYER — COMPREHENSIVE USER GUIDE

Overview & Core Capabilities

biraj-mpv-conf is a clean, high-performance configuration suite for mpv. It brings together a fluent modern interface, tear-free 4K video playback, fast seekbar hover thumbnails, zero-lag track cycling, and native file dialogs into a single seamless experience.

- **Tear-Free Direct3D 11:** Hardware VSync flip presentation (`d3d11-flip=yes`) with low latency.
- **HDR10 & Dolby Vision:** Real-time tone mapping tuned for 0-frame-drop playback on laptop screens.
- **Zero-Lag Track Cycling:** 0ms OSD feedback with 80ms anti-spam debouncing for audio and subtitles.
- **Bounded RAM Memory:** Strict 400 MiB forward and 200 MiB backward buffer capping RAM under 1.0 GB.

Installation & Directory Structure

Installation Mode	Target Directory Path
Standard (Roaming)	<code>C:\Users\<Username>\AppData\Roaming\mpv\</code>
Portable Mode	<code>C:\<PathToMpv>\portable_config\</code>

Video Playback & Tear-Free Pipeline

- **Next-Gen Renderer:** Uses `vo=gpu-next` and `d3d11-flip=yes` to bypass legacy DWM composition.
- **Safe Hardware Acceleration:** Uses `hwdec=auto-safe` with 16 VRAM buffer frames (< 0.5% CPU load).
- **Image Enhancement:** Debanding (`deband=yes`), temporal dithering, and sigmoid upscaling.
- **Subtitle Stall Prevention:** `demuxer-mkv-subtitle-preroll=no` eliminates track switching pauses.

Audio Engine & Zero-Lag Stream Cycling

- **Buffer Headroom:** 500ms audio buffer (`audio-buffer=0.5`) prevents underruns during CPU spikes.
- **DAC Continuity:** `audio-stream-silence=yes` keeps external soundbars and DACs active without delay.
- **Dual Track Shortcuts:** Supports both VLC (`b / v`) and standard MPV (`\_ / ##, j`) cycling.

Memory Architecture & Demuxer Cache Safety

Parameter	Configured Value & Behavior
Forward Buffer	<code>demuxer-max-bytes=400MiB</code> (hard read throttle)
Backward Rewind Buffer	<code>demuxer-max-back-bytes=200MiB</code> (FIFO eviction)
Readahead Window	<code>demuxer-readahead-secs=10</code> (absorbs network jitter)
Peak RAM Footprint	~ 850–950 MB (hard-capped; never climbs indefinitely)
Storage Health	Pure in-memory RAM caching with 0 SSD write wear

Modern UI, Thumbnails & Automation

- **Format Badge** (`hdr_badge.lua`): Auto-announces `DV`, `HDR10+`, `HDR`, `HLG`, `SDR`; toggled with `1`.
- **Seekbar Hover Thumbnails**: `thumbfast` real-time preview thumbnails cached to `\%TEMP\%`.
- **Ascending Playlist Sorter**: `sort_playlist.lua` sorts multi-file drops in numerical order (`k`).
- **Unicode File Pickers**: Native Windows dialogs for media (`Ctrl+o`), subs (`Ctrl+s`), audio (`Ctrl+a`).

Smart Automation Profiles

Profile Name	Trigger Condition & Behavior
<code>[Window-PiP]</code>	Windowed on-top mode: strips window borders and scales OSC by 1.8x
<code>[Minimized]</code>	Auto-pauses playback when the player window is minimized
<code>[Video]</code>	Enables live video playback progress on the Windows taskbar
<code>[Image]</code>	Sets infinite display duration and enables cursor-centered zoom
<code>[night-audio]</code>	Dynamic audio normalizer balancing dialogues and sound effects
<code>[dolby-vision-perf]</code>	Low-overhead tone-mapping profile for smooth 4K DV/HDR playback